49) $w = \sqrt{x^2+y^2+z^2}$

31) $f(x,y) = \sqrt{3x^5y - 7x^3y}$

Sol:-

$$f(x,y) = (3x^5y - 7x^3y)^{1/2}$$

with respect to x

$$= \frac{1}{2}\left[(3x^5y - 7x^3y)^{\frac{1}{2}-1} \frac{\partial}{\partial x}(3x^5y) - \frac{\partial}{\partial x}(7x^3y)\right]$$

$$= \frac{1}{2(3x^5y - 7x^3y)^{1/2}} (15x^4y - 21x^2y)$$

$$= \frac{3x^2y(5x^2 - 7)}{2(3x^5y - 7x^3y)^{1/2}}$$

with respect to y

$$\frac{\partial}{\partial} = \frac{1}{2}(3x^5y - 7x^3y)^{-1/2}\left(\frac{\partial}{\partial y}(3x^5y) - \frac{\partial}{\partial y}(7x^3y)\right)$$

$$= \frac{1}{2}(3x^5y - 7x^3y)^{-1/2}(3x^5 - 7x^3)$$

$$= \frac{1}{2}x^3(3x^2 - 7)(3x^5 - 7x^3y)^{-1/2}$$

3d) $f(x,y) = \frac{x+y}{x-y}$

with respect to x

$$f_x = \frac{(x-y)\frac{\partial}{\partial x}(x+y) - (x+y)\frac{\partial}{\partial x}(x-y)}{(x-y)^2}$$

$$f_x = \frac{(x-y)(1) - (x+y)(1)}{(x-y)^2}$$

$$f_x = \frac{\cancel{(x-y)(x+y)}}{\cancel{y(x-y)^2}} \qquad f_x = \frac{(x-y)-(x+y)}{(x-y)^2}$$

$$f_x = \cancel{x^2-y^2}\ \frac{x+y}{x} \qquad f_x = \frac{x-y-x-y}{(x-y)^2}$$

$$= \frac{-2y}{(x-y)^2}$$

with respect to y

$$= \frac{(x-y)\frac{\partial}{\partial y}(x+y) - (x+y)\frac{\partial}{\partial y}(x-y)}{(x-y)^2}$$

$$= \frac{(x-y)(1) - (x+y)(-1)}{(x-y)^2}$$

$$= \frac{(x-y) + (x+y)}{(x-y)^2}$$

$$= \frac{x-y+x+y}{(x-y)^2}$$

$$= \frac{2x}{(x-y)^2}$$

33) $f(x,y) = y^{-3/2} \tan^{-1}(x/y)$

$$= y^{-3/2}\left(\frac{1}{1+x^2}\right)\frac{\partial}{\partial x}\tan^{-1}\left(\frac{x}{y}\right)$$

$$= y^{-3/2}\cdot\frac{1}{1+\left(\frac{x}{y}\right)^2}\cdot\frac{\partial}{\partial x}\left(\frac{x}{y}\right)$$

$$= y^{-3/2}\cdot\frac{1}{1+\left(\frac{x}{y}\right)^2}\cdot\frac{y+0}{y^2}$$

$$= y^{-3/2}\,\frac{1}{1+\left(\frac{x}{y}\right)^2}\cdot\frac{y}{y^2}$$

$$= \frac{y^{-1/2}}{y^2 + \frac{x^2}{y^2}}$$

$$= y^{-3/2} \cdot \frac{y^2}{y^2 + x^2} \cdot y^{-1}$$

$$= y^{-3/2+1} \cdot \frac{1}{x^2+y^2}$$

$$= y^{-1/2} \cdot \frac{1}{x^2+y^2}$$

$$= \frac{y^{-1/2}}{x^2+y^2}$$

with respect to y

By product rule

$$= \frac{\partial}{\partial y}\left(y^{-3/2}\right) \tan^{-1}\left(\frac{x}{y}\right) + y^{-3/2} \frac{\partial}{\partial y}\left(\tan^{-1}\left(\frac{x}{y}\right)\right)$$

$$= \left(\frac{-3}{2} y^{-5/2}\right) \tan^{-1}\left(\frac{x}{y}\right) + \left(y^{-3/2}\right)\left[\frac{1}{1+\left(\frac{x^2}{y^2}\right)} \frac{\partial}{\partial y}\left(x y^{-1}\right)\right]$$

$$= \left(\frac{-3}{2} y^{-5/2}\right) \tan^{-1}\left(\frac{x}{y}\right) + \left(y^{-3/2}\right)\left[\frac{1}{1+\frac{x^2}{y^2}} \cdot \left(\frac{-x}{y^2}\right)\right.$$

$$= \left(\frac{-3}{2} y^{-5/2}\right) \tan^{-1}\left(\frac{x}{y}\right) + \frac{y^{-3/2} \times y^{\cancel{2}}}{y^2+x^2} - \left(\frac{x}{y^{\cancel{2}}}\right)$$

$$= \frac{-3}{2} y^{-5/2} \tan^{-1}(x/y) + \frac{y^{-3/2}}{y^2+x^2}(-x)$$

$$\frac{-xy^{-3/2}}{y^2+x^2} - \frac{3}{2} y^{-5/2} \tan^{-1} x/y$$

34) $f(x,y) = x^3 e^{-y} + y^3 \sec\sqrt{x}$

with respect to x

$$= \frac{\partial}{\partial x}(x^3) e^{-y} + y^3 \frac{\partial}{\partial x}(\sec\sqrt{x})$$

$$= 3x^2 e^{-y} + y^3 \sec\sqrt{x} \tan\sqrt{x} \frac{\partial}{\partial x}(\sqrt{x})$$

$$= 3x^2 e^{-y} + y^3 \sec\sqrt{x} \tan\sqrt{x} \cdot \frac{1}{2} x^{-\frac{1}{2}}$$

$$= 3x^2 e^{-y} + \frac{x^{-1/2} y^3}{2} \sec\sqrt{x} \sqrt{\tan}\sqrt{x}$$

with respect to y

$$= x^3 e^{-y} \frac{\partial}{\partial y}(-y) + \cancel{3y} \frac{\partial}{\partial y}(y^3) \sec\sqrt{x}$$

$$= -x^3 e^{-y} + 3y^2 \sec\sqrt{x}$$

f3(43)

Find $F(x), F(y), F(z)$

43)

$$f(x,y,z) = z \ln(x^2 y \cos z)$$

with respect to x

$$f_x = z \frac{\partial}{\partial x}\left(\ln x^2 y \cos z\right)$$

$$f_x = z \cdot \frac{1}{x^2 y \cos z} \frac{\partial}{\partial x}\left(x^2 y \cos z\right)$$

$$f_x = z \cdot \frac{1}{(x^2) \cancel{y \cos z}} \cdot 2x \left(\cancel{y \cos z}\right)$$

$$= \frac{2x z}{x^2}$$

$$= \frac{2z}{x}$$

with respect to y

$= z \dfrac{1}{x^2 y \cos z} \cdot x^2 \cos z$

$= \dfrac{z}{y}$

with respect to z
product rule

$= (1) \ln(x^2 y \cos z) + z \cdot \dfrac{1}{x^2 y \cos z} - x^2 y \sin z$

$= \ln(x^2 y \cos z) - z \tan z$

44) $f(x,y,z) = y^{-3/2} \sec\left(\dfrac{xz}{y}\right)$

with respect to x

$fx) = y^{-3/2} \sec\left(\dfrac{xz}{y}\right)$

$= y^{-3/2} \sec\left(\dfrac{xz}{y}\right) \tan\left(\dfrac{xz}{y}\right) \dfrac{\partial}{\partial x}\left(\dfrac{xz}{y}\right)$

$= y^{-3/2} \sec\left(\dfrac{xz}{y}\right) \tan\left(\dfrac{xz}{y}\right) (1)(z)(y^{-1})$

$$= y^{-3/2 - 1} z \sec\left(\frac{xz}{y}\right) \cdot \tan\left(\frac{xz}{y}\right)$$

$$= y^{-5/2} z \cdot \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right)$$

with respect to y

product rule

$$\cancel{f_y = -\frac{3}{2} y^{-3/2 - 1} \left[\sec\left(\frac{xz}{y}\right)\right] + \sec\left(\frac{xz}{y}\right) \frac{\partial}{\partial y}}$$

product rule

$$f_y = \frac{-3}{2} y^{-3/2 - 1} \cdot \sec\left(\frac{xz}{y}\right) + y^{-3/2} \frac{\partial}{\partial y}\left[\sec\left(\frac{xz}{y}\right)\right]$$

$$= \frac{-3}{2} y^{-5/2} \sec\left(\frac{xz}{y}\right) + y^{-3/2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right) \frac{\partial}{\partial y}\left(\frac{xz}{y}\right)$$

$$= \frac{-3}{2} y^{-5/2} \sec\left(\frac{xz}{y}\right) + y^{-3/2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right) \frac{\partial}{\partial y}\left(xzy^{-1}\right)$$

$$= \frac{-3}{2} y^{-5/2} \cdot \sec\left(\frac{xz}{y}\right) + y^{-3/2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right) \; -1xzy^{-2}$$

$$= \frac{-3}{2} y^{-5/2} \cdot \sec\left(\frac{xz}{y}\right) + y^{-3/2-2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right)$$

$$-(xz)$$

$$= \frac{-3}{2} y^{-5/2} \sec\left(\frac{xz}{y}\right) - z y^{-7/2} x \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right)$$

with respect to z

$$= f_z = y^{-3/2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right)$$

$$\frac{\partial}{\partial z}\left(\frac{xz}{y}\right).$$

$$f_z = y^{-3/2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right) \left(\frac{x}{y}\right)$$

$$f_z = y^{-3/2-1} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right) (x)$$

$$f_z = x y^{-5/2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right)$$

45) $f(x,y,z) = \tan^{-1}\left(\frac{1}{xy^2z^3}\right)$

Sol:-

$$= \frac{1}{1+\left(\frac{1}{xy^2z^3}\right)^2} \frac{\partial}{\partial x}\left(\frac{1}{xy^2z^3}\right)$$

$$= \frac{1}{1+\frac{1}{x^2y^4z^6}} \left(-\frac{1}{x^2y^2z^3}\right)$$

$$= \frac{1}{\frac{x^2y^4z^6+1}{x^2y^4z^6}} \left(-\frac{1}{x^2y^2z^3}\right)$$

$$= \frac{\cancel{x^2}y^4z^6}{x^2y^4z^6+1} \left(-\frac{1}{\cancel{x^2}y^2z^3}\right)$$

$$= \frac{-yz^3}{x^2y^4z^6+1} \quad \text{Ans.}$$

with respect to y

$$= \frac{1}{1+\left(\frac{1}{xy^2z^3}\right)^2} \frac{\partial}{\partial y}\left(\frac{1}{xy^2z^3}\right)$$

$$= \left(\frac{1}{1+\frac{1}{x^2y^4z^6}}\right)\left(\frac{-2}{xy^3z^3}\right)$$

$$= \left(\frac{1}{\frac{x^2y^4z^6+1}{x^2y^4z^6}}\right)\left(\frac{-2}{xy^3z^3}\right)$$

$$= \left(\frac{x^2y^4z^6}{x^2y^4z^6+1}\right)\left(\frac{-2}{xy^3z^3}\right)$$

$$= \frac{-2xyz^3}{x^2y^4z^6+1} \quad \text{Ans.}$$

with respect to z

$$= \frac{1}{1+\left(\frac{1}{xy^2z^3}\right)^2}\frac{\partial}{\partial z}\left(\frac{1}{xy^2z^3}\right)$$

$$= \frac{1}{1+\frac{1}{x^2y^4z^6}}\left(\frac{-3}{xy^2z^4}\right)$$

$$= \frac{1}{\frac{x^2y^4z^6+1}{x^2y^4z^6}}\left(\frac{-3}{xy^2z^4}\right)$$

$$= \left(\frac{x^2y^4z^6}{x^2y^4z^6+1}\right)\left(-\frac{3}{xy^2z^4}\right)$$

$$= \frac{-3xy^2z^2}{x^2y^4z^6+1} \quad \text{Ans}$$

47) $w = ye^z \cdot \sin xz$ Find $\frac{\partial w}{\partial x}, \frac{\partial w}{\partial y}, \frac{\partial w}{\partial z}$

Sol:-

w.r.t x

$$\frac{\partial w}{\partial x} = ye^z \frac{\partial}{\partial x}(\sin xz)(z)$$

$$= ye^z (\cos xz)(z)$$

$$= yze^z \cos xz$$

w.r.t y

$$\frac{\partial w}{\partial y} = (1)e^z \sin xz$$

$$= e^z \sin xz$$

w.r.t z

$$\frac{\partial w}{\partial z} = y\frac{\partial}{\partial z}(e^z) + e^z \frac{\partial}{\partial z}(\sin xz)$$

$$= y$$

w.r.t z

$$\frac{\partial w}{\partial z} = ye^z \frac{\partial}{\partial z}(\sin xz) + \sin xz \frac{\partial}{\partial z}(ye^z)$$

$$= ye^z(\cos xz)(x) + ye^z \sin xz$$

$$= ye^z(x\cos xz + \sin xz)$$

48) $w = \frac{x^2 - y^2}{y^2 + z^2}$

Sol:-

w.r.t "x"

$$= \left(\frac{1}{y^2+z^2}\right) \frac{\partial}{\partial x}(x^2 - y^2)$$

$$= \frac{1}{y^2+z^2}(2x)$$

$$= \frac{2x}{y^2+z^2}$$

w.r.t y

$$\frac{\partial w}{\partial y} = \frac{(y^2+z^2)\frac{\partial}{\partial y}(x^2-y^2) - (x^2-y^2)\frac{\partial}{\partial y}(y^2+z^2)}{(y^2+z^2)^2}$$

$$= \frac{(y^2+z^2)(-2y) - (x^2-y^2)(2y)}{(y^2+z^2)^2}$$

$$= \frac{(-2y)(y^2+z^2) - (2y)(x^2-y^2)}{(y^2+z^2)^2}$$

$$= \frac{\cancel{-2y^3} - 2yz^2 - 2yx^2 \cancel{+2y^3}}{(y^2+z^2)^2}$$

$$= \frac{-2y(x^2+z^2)}{(y^2+z^2)^2}$$

w.r.t z

$$= (x^2-y^2)\frac{\partial}{\partial z}(y^2+z^2)^{-1}$$

$$= (x^2-y^2)\left[(-1)(y^2+z^2)^{-2}(2z)\right]$$

$$= \frac{-(x^2-y^2)(2z)}{(y^2+z^2)^2}$$

$$= \frac{2z(y^2 - x^2)}{(y^2+z^2)^2}$$

$$w = \sqrt{x^2+y^2+z^2}$$

Sol:

$$\frac{\partial w}{\partial x} = \frac{\partial}{\partial x}\left(x^2+y^2+z^2\right)^{+\frac{1}{2}}$$

$$= \frac{1}{2}\left(x^2+y^2+z^2\right)^{\frac{1}{2}-1} \frac{\partial}{\partial x}\left(x^2+y^2+z^2\right)$$

$$= \frac{1}{\cancel{2}\left(x^2+y^2+z^2\right)^{\frac{1}{2}}}\left(\cancel{2}x\right)$$

$$= \frac{x}{\sqrt{x^2+y^2+z^2}}$$

w.r.t y

$$\frac{\partial w}{\partial y} = \frac{1}{2}\left(x^2+y^2+z^2\right)^{\frac{1}{2}-1} \frac{\partial}{\partial y}\left(x^2+y^2+z^2\right)$$

$$= \frac{1}{\cancel{2}\left(x^2+y^2+z^2\right)}\left(\cancel{2}y\right)$$

$$= \frac{y}{x^2+y^2+z^2}$$

w.r.t z

Same as above

$$\frac{\partial w}{\partial z} = \frac{z}{x^2+y^2+z^2}$$

Q86

$$f(x,y) = 4x^2 - 8xy^4 + 7y^5 - 3$$

Sol:-

w.r.t x

$f_x = 8x - 8y^4$

w.r.t y

$f_y = -32xy^3 + 35y^4$

Second partial derivative

$f_{xy} = -32y^3$

$f_{yx} = -32y^3$

$f_{xy} = f_{yx}$